

Monochromatic $x+y$ and xy

GREEN-066

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

If $\{1, \dots, N\}$ is r -coloured, then for $N \geq N_0(r)$ there exist integers $x, y \geq 3$ such that $x + y$ and xy have the same colour. Find reasonable bounds for $N_0(r)$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>